

Risk-Aware Two-Way Auctions for AGV Dispatching at an Automated Container Terminal

A discrete-event simulation study

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<https://github.com/Artianwe/agv-terminal-sim>

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Abstract

At an automated container terminal, a quay crane can only hand a container over when an automated guided vehicle (AGV) is underneath it, so the dispatching of AGVs directly limits how fast a ship is worked. I built a discrete-event simulation (Python, SimPy) of the waterside of a terminal with four quay cranes, eight automated stacking cranes and 6–24 AGVs, and compared four dispatching policies: FIFO, nearest vehicle, a centralised Hungarian-assignment controller and a decentralised Contract-Net auction. The basic auction drives the least empty distance but loses up to 3.3 moves per crane-hour when vehicles are scarce and about 1.5 when they are plentiful, because it commits jobs to busy vehicles on the basis of over-confident arrival predictions. I propose a *risk-aware two-way auction*: whichever side is scarce makes the announcement, bids carry a safety margin learned online from past prediction errors, and busy vehicles may only pre-commit when they can be on time and no idle vehicle can. Tuned on separate random seeds and evaluated with common random numbers over 3,800 simulated vessel calls, it is never significantly worse than the best benchmark at any fleet size, is significantly better than nearest vehicle at 14 AGVs (+0.23 moves/QC-h with 13 % less empty travel), and 12 AGVs (three per crane) are enough to reach 90 % of crane capacity.

1 Introduction

Container terminals are multi-machine systems: quay cranes (QCs), horizontal transport and yard cranes must work as one chain, and the slowest link sets the ship’s handling time. In automated terminals the horizontal link is usually a fleet of AGVs. Two design questions follow: *how many AGVs* are needed (fleet sizing), and *how should they be dispatched* (which vehicle takes which container, and when) [1, 2]. Classic dispatching rules are simple and robust [3]; centralised optimisation can do better but is a single point of failure and scales poorly [4, 5]. Multi-agent negotiation, such as the Contract Net Protocol [6], is an attractive middle ground: each machine decides from local information.

This study asks: *can AGVs negotiate jobs among themselves as well as a central controller assigns them, across all fleet sizes?* Section 2 describes the model, Section 3 the policies, Section 4 the experimental design, Section 5 verification and validation, and Section 6 the results.

2 System and model

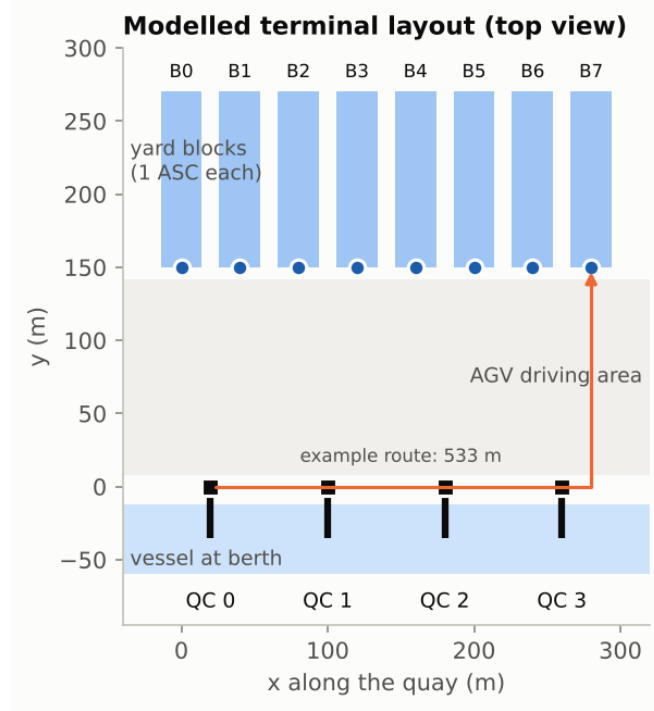


Figure 1: Modelled layout (top view): four QCs on the quay, eight yard blocks with one automated stacking crane (ASC) each, and the AGV driving area in between.

One replication is one vessel call of 1,200 moves ($4 \text{ QCs} \times 300$). Even-numbered QCs discharge first and then load; odd ones do the opposite, which creates chances for an AGV to drop one container at a block and pick up the next one there. Each QC handles its job list strictly in order. A QC cycle is triangular(70, 90, 130) s plus a 10 s hand-over during which an AGV must be underneath; the QC upper bound is therefore 33.8 moves per QC-hour. AGVs drive Manhattan routes $\times 1.3$ at 5 m/s plus 20 s per leg, with lognormal noise (CV 0.1) and a Bureau of Public Roads congestion factor $1 + 0.15 (n_{\text{moving}}/20)^4$. Each ASC is a single server; the AGV waits during the transfer and, for loads, while the ASC fetches the box. Each QC releases its next three jobs to dispatching. The model is a terminating simulation, so no warm-up period is needed. Table 1 lists the parameters and where they come from.

Table 1: Parameters and sources. “Sensitivity” marks values varied in Section 6.4.

Parameter	Value	Source / argument
QCs per vessel	4	as in [8]
QC cycle + hand-over	$\text{tri}(70,90,130) \text{ s} + 10 \text{ s}$	mean 107 s $\Rightarrow$ 33.8 moves/h; Jordan [11]: 72 s computed, ≈ 100 s with yard delays; 22–30 moves/h gross in NW Europe [9]; sensitivity
AGV speed	5 m/s cruise	max. 6 m/s straight, 3 m/s in bends, 2 m/s ² [12]
Overhead per leg	20 s	acceleration/braking, two bends, positioning (10 s in [10])
Route factor	1.3	one-way lane loops vs. straight lines (assumption)
Quay-to-yard depth	150 m	assumption; sensitivity 100–250 m
ASC transfer / store / retrieve	$\text{tri}(20,30,45)$ $\text{tri}(40,60,90)$ $\text{tri}(40,60,90) \text{ s}$	/ ≈ 95 s ASC time per move (≈ 38 waterside moves/h); a / single RMG with landside work does ≈ 30 moves/h [10]; sensitivity $\times 0.8$ –1.25
Travel-time variability	CV 0.10	assumption; sensitivity 0–0.3
Congestion	BPR, $\alpha=0.15$, $\beta=4$, cap. 20	standard BPR coefficients; sensitivity
Jobs released per QC	3	assumption; sensitivity 1–5

3 Dispatching policies

All policies see the same released jobs and must assign each QC’s jobs in sequence order, which rules out deadlock (a unit test exercises deep commitment).

FIFO oldest released job to the AGV that has been idle longest.

Nearest vehicle the idle-AGV/job pair with the shortest empty drive is matched first [3].

Central whenever AGVs are idle, a Hungarian assignment [7] matches all idle AGVs to the released jobs, minimising predicted hand-over time $+ 0.2 \times$ empty drive time.

Basic auction the most urgent QC announces its next job; every AGV with fewer than two jobs bids its own predicted hand-over time $+ 0.2 \times$ empty drive time; the lowest bid wins [6]. Predictions use mean times, the current ASC queue and a bias learned online (exponential moving average of errors).

Risk-aware two-way auction (new) three changes to the basic auction: (i) *two-way*: when released jobs outnumber idle AGVs, the idle AGV announces itself and takes the closest job, as in the availability announcements of the Contract Net Protocol; (ii) *risk margin*: bids use $\max(\hat{t}_{\text{arrive}} + z \hat{\sigma}, t_{\text{need}})$, where $\hat{\sigma}$ is the learned spread of past arrival errors, kept separately for idle and busy AGVs; (iii) *adaptive commitment*: a busy AGV may only win a job if its risk-adjusted arrival is on time and no idle AGV is on time.

4 Experimental design

Each setting is replicated on 10 vessel calls (seeds 1–10). All random times are drawn before the run, so every policy and fleet size sees identical containers and times (common random numbers). We report means with Student- t 95 % confidence intervals and compare policies by *paired* per-seed differences. The two settings of the new auction (z and adaptive commitment) were chosen on separate *tuning* seeds 101–110 with a criterion fixed in advance: the smallest worst-case shortfall (regret) in QC productivity relative to the best benchmark policy at each fleet size (Table 2). The result, $z = 2.5$ with adaptive commitment, was then frozen and evaluated on seeds 1–10.

Table 2: Tuning on seeds 101–110 (best eight of 16 settings plus benchmarks), fleet sizes 6–24.

Setting	Worst-case regret	Mean regret	Empty travel (m/move)
Nearest vehicle	0.08	0.02	137
Risk-aware auction, $z = 2.5$, adaptive commit on	0.12	0.03	133
Risk-aware auction, $z = 3.0$, adaptive commit off	0.13	0.06	125
Risk-aware auction, $z = 3.0$, adaptive commit on	0.13	0.03	133
Risk-aware auction, $z = 2.5$, adaptive commit off	0.17	0.06	122
Risk-aware auction, $z = 2.0$, adaptive commit on	0.20	0.04	133
Risk-aware auction, $z = 2.0$, adaptive commit off	0.21	0.10	120
Risk-aware auction, $z = 1.5$, adaptive commit on	0.27	0.08	134

5 Verification and validation

Verification. 23 automated tests check, among others: a hand-calculated deterministic case (1 QC, 1 AGV: every hand-over time and the 751 s makespan match exactly) for every policy; every job is handled exactly once and in QC sequence; no deadlock with deep commitment; identical work plans across policies (common random numbers); reproducibility; QC productivity never exceeds the crane’s own cycle and reaches it with a very large fleet; AGV time shares sum to one. The full experiment set was run on Linux and on Windows 11 and the summary tables agree to 10^{-14} .

Validation. Simulated QC productivity saturates at ≈ 32.8 moves/QC-h, close to the 34 moves/h Gerrits et al. [8] simulated with 16 AGVs for 4 QCs and above the 22–30 gross moves/h reported for NW European terminals [9], as expected for a model without landside work, breakdowns or bay changes. Our knee of 3–4 AGVs per QC is below the 5.5 per QC suggested in [9]: our yard is close to the quay, traffic is represented only by a congestion function and the QCs are single-hoist. The ranking of policies, not the absolute fleet size, is the robust output.

6 Results

6.1 Fleet sizing

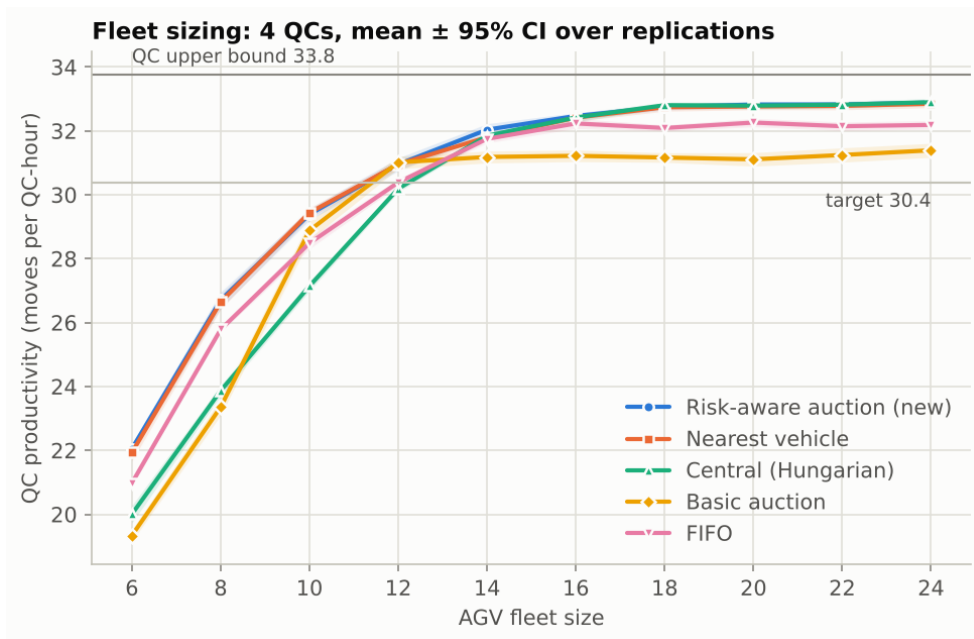
**Figure 2:** QC productivity versus fleet size (mean $\pm$ 95 % CI, 10 vessel calls per point).

Table 3: QC productivity (moves per QC-hour, mean $\pm$ 95 % CI). Bold: best mean at that fleet size.

AGVs	Risk-aware auction	Nearest vehicle	Central (Hungarian)	Basic auction	FIFO
6	22.05 $\pm$ 0.16	21.93 $\pm$ 0.26	20.01 $\pm$ 0.27	19.34 $\pm$ 0.18	20.98 $\pm$ 0.09
8	26.71 $\pm$ 0.25	26.64 $\pm$ 0.22	23.87 $\pm$ 0.16	23.36 $\pm$ 0.25	25.78 $\pm$ 0.08
10	29.39 $\pm$ 0.18	29.44 $\pm$ 0.15	27.14 $\pm$ 0.15	28.88 $\pm$ 0.29	28.49 $\pm$ 0.17
12	30.95 $\pm$ 0.20	30.96 $\pm$ 0.15	30.19 $\pm$ 0.10	31.01 $\pm$ 0.19	30.39 $\pm$ 0.10
14	32.04 $\pm$ 0.18	31.80 $\pm$ 0.11	31.85 $\pm$ 0.08	31.18 $\pm$ 0.19	31.75 $\pm$ 0.12
16	32.47 $\pm$ 0.11	32.41 $\pm$ 0.11	32.42 $\pm$ 0.13	31.22 $\pm$ 0.18	32.23 $\pm$ 0.15
18	32.75 $\pm$ 0.10	32.74 $\pm$ 0.12	32.80 $\pm$ 0.10	31.17 $\pm$ 0.15	32.09 $\pm$ 0.10
20	32.82 $\pm$ 0.10	32.76 $\pm$ 0.10	32.79 $\pm$ 0.11	31.11 $\pm$ 0.21	32.26 $\pm$ 0.13
22	32.83 $\pm$ 0.09	32.78 $\pm$ 0.12	32.82 $\pm$ 0.11	31.25 $\pm$ 0.22	32.15 $\pm$ 0.13
24	32.89 $\pm$ 0.08	32.85 $\pm$ 0.12	32.90 $\pm$ 0.11	31.39 $\pm$ 0.22	32.19 $\pm$ 0.13

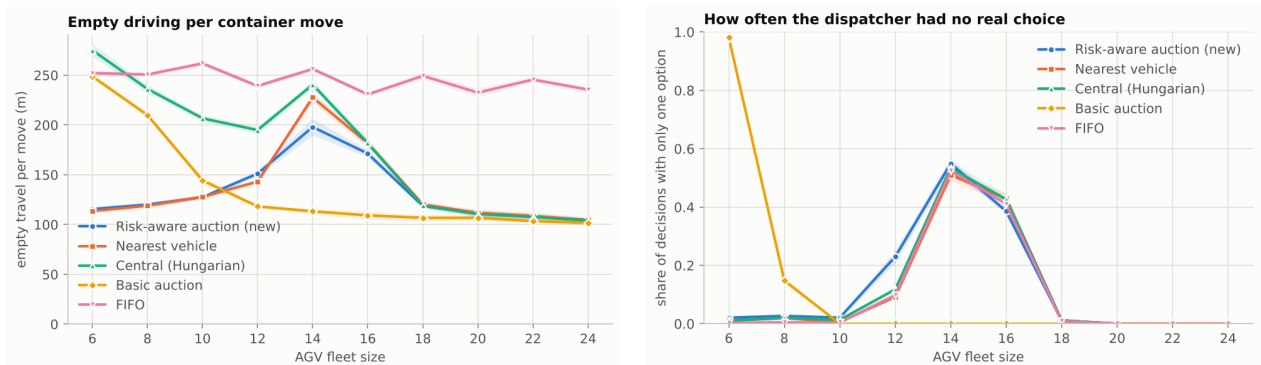
Three regimes appear (Fig. 2, Table 3). With 6–10 AGVs the vehicles are the bottleneck and every metre of empty driving is lost capacity; rules that minimise it (nearest vehicle, the two-way auction) are best, while urgency-driven rules (central, basic auction) lose 2–3 moves/QC-h. With 12–16 AGVs supply roughly matches demand and the policies differ most in how they use the little slack. With 18 or more AGVs every policy except FIFO and the basic auction reaches ≈ 32.8 moves/QC-h. For a 90 % target (30.4 moves/QC-h), 12 AGVs (three per QC) suffice with the new auction, nearest vehicle or the basic auction, but the central controller needs 14; for 95 % all policies except the basic auction need 16.

Table 4: Risk-aware auction minus each reference, paired over the same 10 seeds (mean $\pm$ 95 % CI). Δ prod. in moves/QC-h; Δ empty in m per move (negative = less empty driving). Bold: significant.

AGVs	vs nearest vehicle		vs central		vs basic auction	
	Δ prod.	Δ empty (m)	Δ prod.	Δ empty (m)	Δ prod.	Δ empty (m)
6	+0.12 $\pm$ 0.28	+2 $\pm$ 3	+2.05 $\pm$ 0.32	-159 $\pm$ 6	+2.72 $\pm$ 0.28	-133 $\pm$ 3
8	+0.06 $\pm$ 0.23	+1 $\pm$ 3	+2.83 $\pm$ 0.24	-116 $\pm$ 4	+3.35 $\pm$ 0.35	-90 $\pm$ 4
10	-0.05 $\pm$ 0.16	+0 $\pm$ 3	+2.24 $\pm$ 0.20	-79 $\pm$ 3	+0.51 $\pm$ 0.35	-17 $\pm$ 4
12	+0.00 $\pm$ 0.16	+8 $\pm$ 4	+0.76 $\pm$ 0.14	-44 $\pm$ 2	-0.06 $\pm$ 0.20	+33 $\pm$ 4
14	+0.23 $\pm$ 0.14	-30 $\pm$ 11	+0.19 $\pm$ 0.19	-43 $\pm$ 11	+0.85 $\pm$ 0.19	+85 $\pm$ 9
16	+0.05 $\pm$ 0.13	-10 $\pm$ 6	+0.05 $\pm$ 0.09	-10 $\pm$ 5	+1.25 $\pm$ 0.17	+62 $\pm$ 4
18	+0.01 $\pm$ 0.10	-2 $\pm$ 3	-0.05 $\pm$ 0.07	+0 $\pm$ 3	+1.58 $\pm$ 0.15	+12 $\pm$ 3
20	+0.06 $\pm$ 0.07	+0 $\pm$ 3	+0.03 $\pm$ 0.07	+1 $\pm$ 4	+1.71 $\pm$ 0.23	+5 $\pm$ 4
22	+0.04 $\pm$ 0.10	-1 $\pm$ 3	+0.01 $\pm$ 0.07	+0 $\pm$ 3	+1.58 $\pm$ 0.21	+4 $\pm$ 2
24	+0.04 $\pm$ 0.12	+0 $\pm$ 2	-0.01 $\pm$ 0.06	+1 $\pm$ 2	+1.49 $\pm$ 0.22	+3 $\pm$ 4

Table 4 is the main result. The risk-aware auction is never significantly worse than any benchmark. Against nearest vehicle, the strongest benchmark, it is significantly better at 14 AGVs (+0.23 moves/QC-h and 30 m less empty driving per move) and equal elsewhere; against the central controller it gains 0.8–2.8 moves/QC-h up to 12 AGVs; against the basic auction it gains 0.5–3.3 moves/QC-h at every fleet size except 12, at the price of more empty driving in the 12–16 range.

6.2 Why empty driving peaks at 14 AGVs

**Figure 3:** Left: empty travel per move. Right: share of dispatch decisions where only one option existed.

For idle-only policies, empty driving is not monotonic in fleet size: it peaks at 14 AGVs (Fig. 3). The right panel explains why: at that size supply $\approx$ demand, and about half of all decisions have exactly one idle AGV and one job, so the dispatcher has no choice. With fewer AGVs, a freed AGV can choose among up to four released jobs; with more, a job can choose among many idle AGVs.

6.3 Ablation

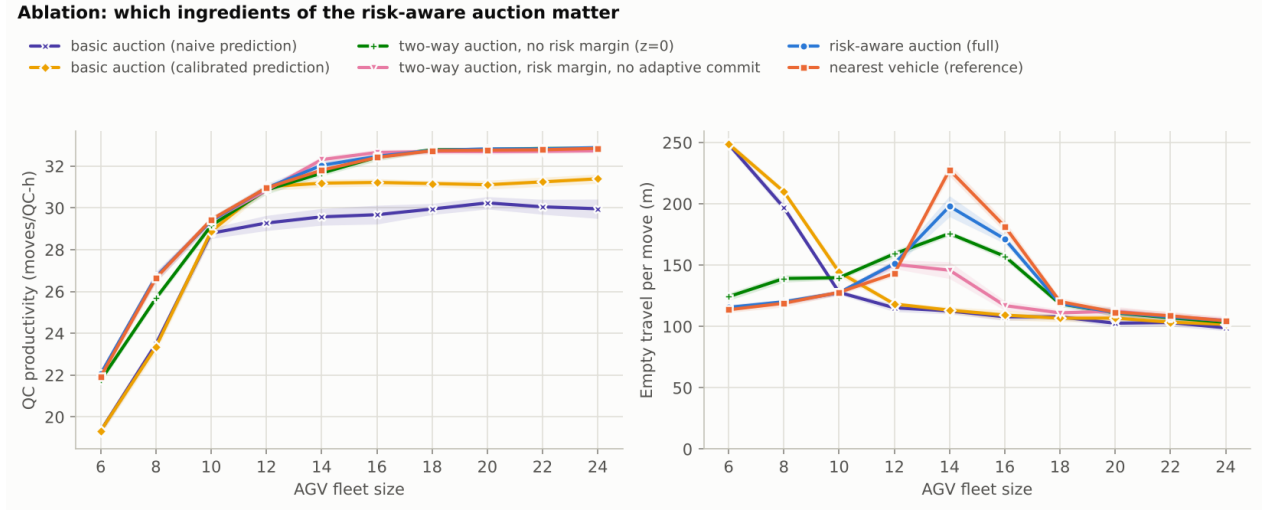


Figure 4: Ablation: switching on the ingredients of the risk-aware auction one at a time.

Calibrated predictions alone lift the basic auction by 0.9–1.6 moves/QC-h at 14–24 AGVs. The two-way rule, together with the on-time condition for busy AGVs, removes most of the remaining losses at both ends (+2.4 moves/QC-h at 6 AGVs, +1.4 to +1.7 at 18–24). The learned risk margin adds a further 0.1–1.0 moves/QC-h at 6–14 AGVs. Without adaptive commitment the auction drives less empty distance and is even slightly faster at 14–16 AGVs, but up to 0.16 moves/QC-h slower with 20 or more: a genuine efficiency–productivity trade-off, which the worst-case tuning criterion resolved in favour of adaptive commitment (Fig. 4).

6.4 Sensitivity

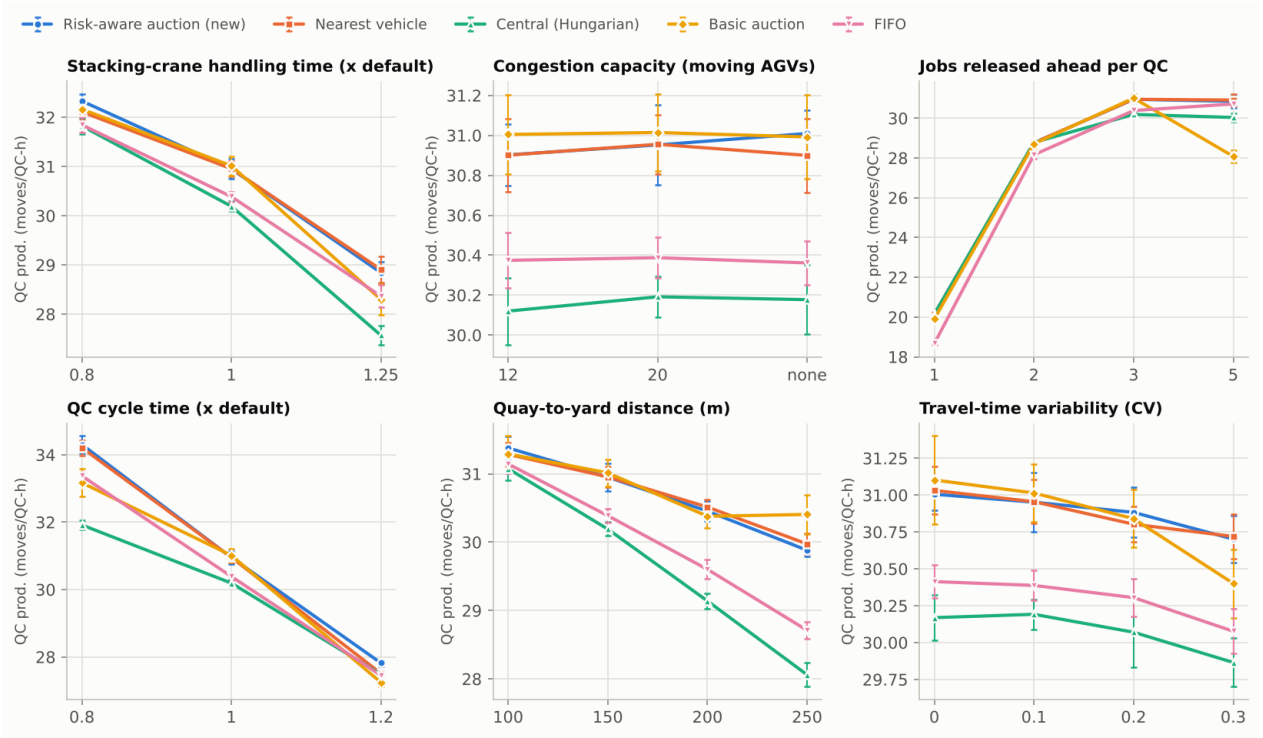


Figure 5: One-factor-at-a-time sensitivity at 12 AGVs.

The new auction stays within the confidence band of the best policy for every factor except the deepest yard (250 m), where the basic auction is about 0.5 moves/QC-h ahead (Fig. 5). The basic auction is fragile when more jobs are released ahead (lookahead 5: -2.9 moves/QC-h) and when travel times become more uncertain (CV 0.3: -0.7); the risk-aware version loses only -0.1 and -0.3 . The central controller degrades fastest with yard depth, because its urgency-first assignment costs empty driving.

Table 5: Policy comparison at 12 AGVs (mean $\pm$ 95 % CI).

Policy	QC prod. (moves/QC-h)	Empty travel (m/move)	Crane wait (s/move)	Messages/job
Risk-aware auction	30.95 ± 0.20	151 ± 3	9.6 ± 0.7	31
Nearest vehicle	30.96 ± 0.15	143 ± 2	9.6 ± 0.4	–
Central (Hungarian)	30.19 ± 0.10	195 ± 4	12.6 ± 0.4	–
Basic auction	31.01 ± 0.19	118 ± 2	9.4 ± 0.8	10
FIFO	30.39 ± 0.10	239 ± 2	11.8 ± 0.3	–

7 Discussion and limitations

A decentralised auction can match a centralised controller and a strong greedy rule *if* the agents know how uncertain their own predictions are and the protocol lets the scarce side speak first. Its cost is communication: about 31 messages per container at 12 AGVs, roughly three times the basic auction (Table 5). Limitations: single-load AGVs; no explicit lanes, collision avoidance or traffic deadlocks; random yard-block allocation; no battery charging; no buffer lanes or lift-AGVs at the crane; one vessel and no landside traffic; ASCs do not pre-fetch; several parameters are assumptions (Table 1). Natural next steps are an explicit lane network, energy-aware dispatching and testing the protocol under communication delays and vehicle failures, the situations where decentralisation is meant to pay off.

8 Conclusion

Within this model, 12 AGVs (three per quay crane) give 90 % of crane capacity. A basic Contract-Net auction is efficient but brittle. Adding a two-way announcement rule, learned risk margins and adaptive commitment makes it as good as the best of four benchmarks at every fleet size from 6 to 24 and better in the critical balanced regime, without a central controller.

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